

Quiz (Carolina Reaper)

- Q1.** A shipping company has a quadratic equation $x^2 + 5x + 6 = 0$ to model its delivery times. What is the discriminant?
(A) 1
(B) 13
(C) -11
(D) 25
(E) -1
- Q2.** For the quadratic equation $x^2 - 4x - 3 = 0$, how many solutions are there?
(A) One
(B) Two
(C) None
(D) Infinite
(E) Depends on the value of x
- Q3.** A traffic flow model is given by $x^2 - 2x - 15 = 0$. What are the solutions?
(A) $x = -3, x = 5$
(B) $x = 3, x = -5$
(C) $x = 1, x = -15$
(D) $x = -1, x = 15$
(E) $x = 0, x = -15$
- Q4.** The quadratic formula for solving $ax^2 + bx + c = 0$ is $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. What is the purpose of $b^2 - 4ac$?
(A) To find the value of a
(B) To find the value of b
(C) To determine the number of solutions
(D) To determine the value of c
(E) To find the solution to the equation
- Q5.** A package delivery company uses the equation $x^2 + 2x + 1 = 0$ to model its delivery route. What is the discriminant and how many solutions are there?
(A) Discriminant is 0, one solution
(B) Discriminant is 4, two solutions
(C) Discriminant is -4, no solutions
(D) Discriminant is 1, one solution
(E) Discriminant is -1, no solutions
- Q6.** For the equation $x^2 - 9x + 20 = 0$, what is the value of the discriminant?
(A) 41
(B) 49
(C) 1
(D) -11
(E) 81
- Q7.** The equation $x^2 + 4x + 4 = 0$ has how many solutions?
(A) One
(B) Two
(C) None
(D) Infinite
(E) Depends on the value of x
- Q8.** A logistics company uses the quadratic equation $x^2 - 7x + 12 = 0$ to model its shipping routes. What is the discriminant?
(A) 1
(B) 13
(C) 25
(D) 49
(E) -11

Open Ended Questions (FRQ)

- Q9.** Solve the quadratic equation $x^2 - 6x + 8 = 0$ using the quadratic formula. Show all steps and determine the number and type of solutions.
- Q10.** A transportation company uses the equation $x^2 + 3x - 10 = 0$ to model its traffic flow. Determine the discriminant, solve for x , and describe the solutions in the context of the problem.

Chef's Tips

- Remind students to check their work and use the correct formula
- Emphasize the importance of determining the discriminant to find the number of solutions
- Encourage students to use real-world examples to illustrate their answers